MTH104 - Sets and Logic (Quiz 3)

Question # 2 of 10 (Start time: 03:40:52 PM, 21 June 2025)

The set of positive integers $P = \{1, 2, 3, \ldots\}$ is not order-isomorphic to the set of even positive integers $E = \{2, 4, 6, \ldots\}$.

Select the correct option

○	True
◉	False

MTH104 - Sets and Logic (Quiz 3)

Question # 1 of 10 (Start time: 03:40:34 PM, 21 June 2025)

A is bounded if _____.

Select the correct option

○	A has only a lower bound.
●	A has an upper and lower bound.
○	None of these.
○	A has only an upper bound.

MTH104 - Sets and Logic (Quiz 3)

Question # 6 of 10 (Start time: 03:41:43 PM, 21 June 2025)

If a lower bound of A precdes every other lower bound of A, then it is called the infimum of A

Select the correct option

◉	Given statement is false.
○	Given statement is true.

MTH104 - Sets and Logic (Quiz 3)

Question # 3 of 10 (Start time: 03:41:05 PM, 21 June 2025)

Let A be a well-ordered set. The initial segment s(a) of an element $a \in A$ consists of all elements in A which strictly precedes a.

Select the correct option

◉	Given statement is true.
○	Given statement is false.

MTH104 - Sets and Logic (Quiz 3)

Question # 8 of 10 (Start time: 03:42:05 PM, 21 June 2025)

Let A be an ordered set. Then A is said to be well-ordered if every subset of A contains a first element.

Select the correct option

- (●) Given statement is true.
- ( ) Given statement is false.

MTH104 - Sets and Logic (Quiz 3) **Quiz Start Time: 03:40 PM, 21 June 2025**

Question # 5 of 10 (Start time: 03:41:31 PM, 21 June 2025) **Total Marks:** 1

Let S be a partially ordered set, and let A be a subset of S. An element n in S is called a lower bound of A if n precedes every element of A, that is, for every $y \in A$, we have $y \geq n$.

Select the correct option

Reload Math Equations

- (●) Given statement is true.
- ( ) Given statement is false.

MTH104 - Sets and Logic (Quiz 3) Quiz Start Time: 03:40 PM, 21 June 2025

Question # 4 of 10 (Start time: 03:41:18 PM, 21 June 2025) **Total Marks:** 1

Let S be a partially ordered set, and let A be a subset of S. If A has a lower bound we say A is bounded below.

Select the correct option

Reload Math Equations

- ○ Given statement is false.
- ◉ Given statement is true.

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz 3) Quiz Start Time: 03:40 PM, 21 June 2025

Question # 9 of 10 (Start time: 03:42:19 PM, 21 June 2025) **Total Marks:** 1

Let S be a partially ordered set, and let A be a subset of S. An element m in S is called an upper bound of A if m succeeds every element of A, that is, for every $x \in A$, we have $x \leq m$.

Select the correct option

Reload Math Equations

- ○ Given statement is false.
- ◉ Given statement is true.

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz 3) Quiz Start Time: 03:40 PM, 21 June 2025

Question # 7 of 10 (Start time: 03:41:54 PM, 21 June 2025) **Total Marks:** 1

Two ordered sets X and Y are said to be order-isomorphic or isomorphic or similar if there exists a one-to-one correspondence $f : X \rightarrow Y$ which preserves the order relations. Such a function f is then called an order-isomorphism or isomorphism from X onto Y.

Select the correct option

Reload Math Equations

- ◉ Given statement is true.
- ○ Given statement is false.

Click to Save Answer & Move to Next Question

MTH104 - Sets and Logic (Quiz 3) Quiz Start Time: 03:40 PM, 21 June 2025

Question # 10 of 10 (Start time: 03:42:42 PM, 21 June 2025) **Total Marks:** 1

Suppose X and Y are partially ordered sets. A one-to-one(injective) function $f: X \rightarrow Y$ is called a similar mapping from X into Y if f preserves the order relation, that is, if the following conditions holds for any pair $a, b \in X$: $a \leq b$ in X if and only if $f(a) \leq f(b)$ in Y.

Select the correct option

Reload Math Equations

- ○ Given statement is false.
- ◉ Given statement is true.

Click to Save Answer & Move to Next Question